Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2024-P1-Model)** **Date: 29-04-2026**

Time: 09.00Am to 12.00Pm **GTA-28** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**B**	2	**D**	3	**D**	4	**C**	5	**BC**	6	**AC**
7	**ABC**	8	**712**	9	**25**	10	**6**	11	**3**	12	**5**
13	**19**	14	**B**	15	**A**	16	**B**	17	**A**		

PHYSICS

18	**A**	19	**D**	20	**A**	21	**A**	22	**BC**	23	**CD**
24	**CD**	25	**6**	26	**2**	27	**6**	28	**2**	29	**5**
30	**3**	31	**B**	32	**B**	33	**B**	34	**A**		

CHEMISTRY

35	**B**	36	**C**	37	**B**	38	**C**	39	**B**	40	**AC**
41	**ABCD**	42	**8**	43	**4**	44	**20**	45	**3**	46	**286**
47	**4**	48	**C**	49	**D**	50	**A**	51	**C**		

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SOLUTIONS
MATHEMATICS

1. We can rewrite $(1-2i)z+(-2i-1)\bar{z}=6i$ as $\frac{z-\bar{z}}{2i}=z+\bar{z}+3$. If we let

z = x + yi, this is equivalent to the equation y = 2x + 3. Suppose that a point $u=\frac{1}{z}$

where $(1-2i)z+(-2i-1)\bar{z}=6i$. Let u = wi. Then, $z=\frac{1}{u}=\frac{v}{v^2+w^2}-\frac{w}{v^2+w^2}$ i

and we must also have - $-\frac{w}{v^2+w^2}=2\frac{v}{v^2+w^2}+3$

This can be rewritten as $\left(v+\frac{1}{3}\right)^2+\left(w+\frac{1}{6}\right)=\frac{5}{36}$

Note that we cannot allow $v^2+w^2=0$ but the origin is still included in the set of points we consider given the problem statement. The area of the circle described by this equation is $\frac{5\pi}{36}$

2. $$\frac{P\left(S_1\cap\left(\overline{S_2\cup S_3}\right)\right)}{p\left(\overline{S_2\cup S_3}\right)}=\frac{1}{10}$$

$$\Rightarrow P\left(S_1\cap\left(\overline{S_2\cup S_3}\right)\right)+P\left(S_2\cup S_3\right)=\frac{1}{2}$$

$$\frac{1}{2}-P\left(S_2\cup S_3\right)=\frac{1}{10}\left(1-P\left(S_2\cup S_3\right)\right)$$

$$P\left(S_2\cup S_3\right)\left(1-\frac{1}{10}\right)=\frac{1}{2}-\frac{1}{10}\Rightarrow P(S_2\cup S_3)=\frac{10}{9}\cdot\frac{8}{20}$$

P(W) + P (T) = P $(S_2\cup S_3)$

$$P(T)=P(S_2\cup S_3)-P(W)$$

$$\frac{4}{9}-\frac{1}{12}=\frac{39}{9\times12}=\frac{13}{36}$$

3. (A) RHD at x = 0 : $\lim\limits_{h\to0}\dfrac{\left(2-2h^2-h^2\sin\frac{1}{h}\right)-2}{h}=0$

Similarly LHD at x = 0 is also equal to 0.
Differentiable at x = 0

(B) $f'(x)=-4x-2x\sin\frac{1}{x}-x^2\left(\cos\frac{1}{x}\right)\left(\frac{-1}{x^2}\right)$

$$f'(x)=\left(4x+2x\sin\frac{1}{x}\right)+\cos\frac{1}{x}$$

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$$f'(x)=-\left(2x\left(2-\sin\frac{1}{x}\right)\right)+\cos\frac{1}{x}$$

for $x\in(0,\delta)$

$\Rightarrow$We can't say f(x) is decreasing on $(0,\delta)$ as $\cos\frac{1}{x}$ oscillates.

(C) for $x\in(-\delta,0)$, for any $\delta>0$

$\Rightarrow$f(x) is not increasing on $(-\delta,0)$ as $\cos\frac{1}{x}$ oscillates from -1 to 1.

D) f(0) = 2
f(0 + h) < 2
f(0 – h) < 2
x = 0 is local maxima

4. $f(x+1)=x+f(x)$

$e^{f(x+1)}=e^{x+f(x)}=e^{x}.e^{f(x)}$

$g(x+1)=e^{x}g(x)$

$\ell ng(x+1)=x+\ell ng(x)$

$\frac{g'(x+1)}{g(x+1)}=1+\frac{g'(x)}{g(x)}$

$\therefore\frac{g'(x+1)}{g(x+1)}-\frac{g'(x)}{g(x)}=1$

Put x = $\frac{1}{2},\frac{g'\left(\frac{1}{2}+1\right)}{g\left(\frac{1}{2}+1\right)}-\frac{g'\left(\frac{1}{2}\right)}{g\left(\frac{1}{2}\right)}=1$

Putx = $\frac{3}{2},\frac{g'\left(\frac{1}{2}+2\right)}{g\left(\frac{1}{2}+2\right)}-\frac{g'\left(\frac{3}{2}\right)}{g\left(\frac{3}{2}\right)}=1$

5. This comes down to computing the maximum of $\frac{1+7s}{2+s}=7-\frac{13}{2+s}$, where s is non – negative number denoting $\left(\frac{b}{a}\right)^2$. In order to maximize this expression, $\frac{13}{2+s}$ should be minimized, i.e $\left|\frac{b}{a}\right|$ be chosen as large as possible.

The value of $\frac{b}{a}$ is equal to the slope of the line passing through the origin and (a, b). The line with the greatest slope that passes through the origin and meets the circle must be tangent to the circle. Drawing tangent lines through the origin to the circle, one sees that the point (a, b) we desire lies at the vertex of the right angle of a right triangle with

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hypotenuse 5 and a leg of length 4. We find that (a, b) = $\left(\frac{9}{5},\frac{12}{5}\right)$, so $\frac{b}{\alpha}=\frac{4}{3}$. Plugging this into the expression, we get $7-\frac{13}{2+\frac{16}{9}}=\frac{121}{34}$

6. The probability that the value of the minimum element is i is $\left(\frac{6-i}{5}\right)-\left(\frac{5-i}{5}\right)^5$ since $\left(\frac{6-i}{5}\right)^5$ is the probability that all the integers selected are at least I and we must subtract the probability $\left(\frac{5-i}{5}\right)^5$ that all the integers selected are at least i + 1. This us

$$\sum_{i=1}^{5} i\left(\left(\frac{6-i}{5}\right)^5-\left(\frac{5-i}{5}\right)^5\right)=\sum_{i=1}^{5}\left(\frac{i}{5}\right)^5$$

$$=\frac{177}{125}$$

7. $\bar{a}\times\bar{b}=\bar{c}\Rightarrow\left(\bar{a}\times\bar{b}\right)\cdot\bar{c}=\bar{c}\cdot\bar{c}$

$\therefore\left[\bar{a}\bar{b}\bar{c}\right]=|\bar{c}|^2$ …(i)

and $\left(\vec{b}\times\vec{c}\right)=\vec{a}\Rightarrow\bar{a}\cdot(\bar{b}\times\bar{c})=\vec{a}.\bar{c}$

$\therefore[\bar{a}\bar{b}\bar{c}=|\bar{a}|^2=4$ …(ii)

$\therefore|\bar{a}|=|\bar{c}|=2$

Option : (1)

$\bar{a}\times\left(\left(\bar{b}+\bar{c}\right)\times\left(\bar{b}-\bar{c}\right)\right)=\bar{a}\times\left(-\bar{b}\times\bar{c}+\bar{c}\times\bar{b}\right)$

$=2\bar{a}\times(\bar{c}\times\bar{b})=2\bar{a}\times(\bar{a})=0$

Option: (2)

Projection of $\bar{a}$ on $\left(\bar{b}\times\bar{c}\right)=\frac{\bar{a}\cdot(b\times\bar{c})}{|b\times c|}=\frac{|\bar{a}|^2}{|a|}=2$

Option: (3)

$\left[\bar{a}\bar{b}\bar{c}\right]+\left[\bar{c}\bar{a}\bar{b}\right]=2\left[\bar{a}\bar{b}\bar{c}\right]=8$

Option : (4)

$|3\bar{a}+\bar{b}-2\bar{c}|^2$

$=9\bar{a}^2+\bar{b}^2+4\bar{c}^2+6\bar{a}.\bar{b}-4\bar{b}\cdot\bar{c}-12\bar{a}\cdot\bar{c}$

$=53$

$\because\bar{a}=\bar{b}\cdot\bar{c}=\bar{c}\cdot\bar{a}=0$

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and $\left[\bar{a}\bar{b}\bar{c}\right]^2 = \begin{vmatrix} \bar{a}\cdot\bar{a} & \bar{a}\cdot\bar{b} & \bar{a}\cdot\bar{c} \\ \bar{b}\cdot\bar{a} & \bar{b}\cdot\bar{b} & \bar{b}\cdot\bar{c} \\ \bar{c}\cdot\bar{a} & \bar{c}\cdot\bar{b} & \bar{c}\cdot\bar{c} \end{vmatrix}$

$\therefore 16 = |\bar{a}|^2|\bar{b}|^2|\bar{c}|^2$

$\therefore |\bar{b}| = 1$

8. Suppose that the first term of the sequence ΔA is d then

$\Delta A = \{d, d+1, d+2 \ldots\ldots (d+(n-1)\}$

Hence A = $(a_1, a_1+d, a_1+d+(d+1)$

$a_1+d+(d+1)+(d+2)\ldots..)$

$a_n = a_1+(n-1)d+\frac{1}{2}(n-1)(n-2)$

So $a_n = \frac{(n-\alpha)(n-\beta)}{2}$

Since $a_{19} = a_{92} = 0$ we must have

$a_n = \frac{1}{2}(n-19)(n-92)$

So $a_3 = \frac{1}{2}(3-19)(3-92) = 712$

9. $\sum_{i=0}^{k} {}^{10}C_1 \cdot {}^{15}C_{k-i} + \sum_{i=0}^{k+1} {}^{12}C_1 \cdot {}^{13}C_{k+1-i}$

$= {}^{25}C_k + {}^{25}C_{k+1}$

$= {}^{26}C_{k+1}$

$0 \le k+1 \le 25$

$-1 \le k \le 24$

But ${}^{13}C_{k+1-i}$ exists for

then $0 \le i \le k+1$

$\Rightarrow k \le 12$

Hence $k_{max} = 12$

10. For consistency

$\begin{vmatrix} a & 1 & b \\ b & 1 & a \\ a & b & ab \end{vmatrix} = 0$

$\Rightarrow (a-b)(a-b^2) = 0$

$\Rightarrow$Either a = b or $a = b^2$

When a = b, then ordered pairs are (0, 0), (1, 1), (2, 2), (3, 3), (4, 4)

When $a = b^2$, then (4, 2)

Hence, Number of ordered pairs are 6.

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11. $f(x)=\prod_{K=1}^{\infty}\left(\frac{1+2\cos\left(\frac{2x}{3^K}\right)}{3}\right)=\prod_{K=1}^{\infty}\left(\frac{1+2\left(1-2\sin^2\frac{x}{3^K}\right)}{3}\right)$

$f(x)=\prod_{K=1}^{\infty}\left(\frac{3-4\sin^2\frac{x}{3^K}}{3}\right)=\prod_{K=1}^{\infty}\left(\frac{3\sin\frac{x}{3^K}-4\sin^3\frac{x}{3^K}}{3\sin\frac{x}{3^x}}\right)$

$f(x)=\prod_{K=1}^{\infty}\left(\frac{\sin\frac{3x}{3^K}}{3\sin\frac{x}{3^K}}\right)=\lim_{n\to\infty}\prod_{K=1}^{\infty}\left(\frac{\sin\frac{x}{3^{K-1}}}{3\cdot\sin\frac{x}{3^k}}\right)$

$=\lim_{n\to\infty}\left(\frac{\sin^x}{3\sin\frac{x}{3}}\cdot\frac{\sin\frac{x}{3}}{3\cdot\sin\frac{x}{3^2}}.....\frac{\sin\frac{x}{3^{n-1}}}{3\sin\frac{x}{3^b}}\right)$

$\therefore f(x)=\frac{\sin x}{x};x\neq 0$

From (1) f(0) = 1

$\therefore f(x)=\begin{cases}\frac{\sin x}{x} & :x\neq 0\\ 1 & :x=0\end{cases}$

$[xf(x)]+|xf(x)|+(x-1)|x^2-3x+2|$ is

12. Let S = {1, 2, 3, 4, 5, 6, 7}. Note that $S_i\cap S_j=S_i$ is equivalent to $S_i\subseteq S_j$. We use constructive counting and select S_5first, and then move down to S_1. First, note that there are $\binom{7}{5}$ ways to choose S_5. Now, S_4must be a subset of S_5(as if $S_4\cap S_5=\theta$ then $|S|\geq 4+5=9$), so there are $\binom{5}{4}$ ways to choose S_4. Moreover, if $S_3$is not a subset of S_4, then $S_3=S-S_4$. But then $S_3\cap S_5\neq S_3$ and $S_3\cap S_5\neq\theta$, a contradiction. Thus $S_3\subseteq S_4\subseteq S_5$ and it follows that there are $\binom{4}{3}$ ways to choose S_3. Now, either $S_2$is a subset of S_3, or $S_2\cap S_3=\theta$. In the first case, there are $\binom{3}{2}$ ways to choose S_2. In the latter case, since $S_2$is not a subset of S_3, if cannot be a subset of S_4 and S_5. It follows that

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$S_2 \cap S_4, S_2 \cap S_5 = \theta$ and $S_2 = S - S_5$. Thus, there are $\binom{3}{2}+1$ total ways to select S_2.

Finally, note that S_1 can be any of {1, 2, 3, 4, 5, 6, 7}. The answer is

$$\binom{7}{5}\binom{5}{4}\binom{4}{3}\left(\binom{3}{2}+1\right)7$$
$$= 21 \cdot 5 \cdot 4 \cdot 4 \cdot 7 = 11760$$

13. From left to right, let A be the event that the first there students are seated in increasing age order, let B be the event that the middle three students are seated in increasing age order, and let C be the event that the last three students are seated in increasing age order. Using PIE, |A|: Choose 3 students to seat in increasing age order. Seat the last two students in any order. There are $\binom{5}{3} * 2! = 20$ ways to do this. $|B|$ and $|C|$ both follow the same logic, yielding 20 ways each. $|A \cap B|$: this counts the number of ways to seat the first 4 students from left to right in increasing age order. This $\binom{5}{4} * 1 = 5$ $|B \cap C|$ is identical to $|A \cap B|$, yielding 5 ways. $|A \cap C|$ is the number of ways to seat all 5 students in order, yielding 1 way. $|A \cap B \cap C|$ also the number of ways to seat all 5 students in order, yielding 1 way. In total, we get
$|A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap B \cap C| = 20 + 20 + 20 - 5 - 5 - 1 + 1 = 50$ ways to seat the students in increasing age order.
Then the number of ways to not seat students in increasing order is
$5! - 50 = 120 - 50 = 70$. The probability that no three consecutive students are seated in increasing age order is then $\frac{70}{120} = \frac{7}{12}$. Finally, 7 + 12 = 19

14. A) Solving $y = \frac{2K^2 - K}{3} > 0$ and $x = \frac{4K - 2K^2}{3} > 0 \Rightarrow K \in \left(\frac{1}{2}, 2\right)$

B) Given g(x) = $\ln\left(\cos^{-1} x\right)$; As $0 \le \cos^{-1} x \le \pi \forall x[-1,1]$

So domain of g(x) = [-1, 1).
Hence number of integers are two (i.e. -1 and 0)
(C) Clearly, domain of expression = {-1, 1}
As x > 0 given
So x = 1
Hence value of expression is 1

(D) Here $f(x) = \frac{x+a}{x+b}$ …(1)

And $f^{-1}(x) = \frac{a - bx}{x - 1}$ …(2)

Given f(x) = $f^1(x) \Rightarrow \frac{x+a}{x+b} = \frac{a-bx}{x-1}$

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$\Rightarrow (1+b)x^2+(b^2-1)x-x(1+b)=0\forall x\in D_f$

Hence b = -1 and $a\in R$

15. $f(x)=\begin{cases}-2; & x\in[-2,1)\\ -1; & x\in[-1,0)\\ x; & x\in[0,2]\end{cases}$ and g(x) = sec x; $x\in R-(2n+1)\frac{\pi}{2}, n\in I$

$fog\begin{cases}-2; & \sec x\in[-2,-1)\\ -1; & \sec x\in[-1,0)\\ \sec x; & \sec x\in[0,2]\end{cases}$

We have

$fog(x)=\begin{cases}-2; & x\in\left[-\frac{4\pi}{3},-\frac{2\pi}{3}\right]\cup\left[\frac{2\pi}{3},\frac{4\pi}{3}\right]-\{-\pi,\pi\}\\ -1; & X=-\pi,\pi\\ \sec x; & X\in\left[-\frac{\pi}{3},\frac{\pi}{3}\right]\end{cases}$

Limit of fog exist at x = $-\pi,\pi-1$, points of discontinuity of fog are $-\pi,\pi$ and points of Differentiability of fog are – 1, $\frac{5\pi}{6}$.

$g(f(x))=\begin{cases}\sec(-2); & x\in[-2,-1)-\left\{-\frac{\pi}{2}\right\}\\ \sec(-1); & x\in[-1,0)\\ \sec x; & x\in[0,2]-\left\{\frac{\pi}{2}\right\}\end{cases}$

Limit of g(f(x)) does not exist at x =-1

16. (P) $\cos\left|x-k+\frac{1}{2}\right|$ is differentiable everywhere

$\Rightarrow \sin|x-k|$ is NOT diff. at k = 0,1,2,3,4

(Q) P(x) = $10x^3-45x^2+60x+35$

$P'(x)=30(x-1)(x-2)$

$P(x)$ decreases in [1, 2]

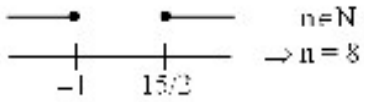
$\Rightarrow$Range of P(x) = [55,60]

$f(x)=\left[\frac{p(x)}{n}\right]$ min value of n = 9

(R) For g(x) to be increasing

$2n^2-13n-15\geq 0$

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(S) $h(x)=(x^2-9)^n(x^2+2x+3)$

$h'(x)=(x^2-9)^n(2x+2)+(x^2+2x+3)n(x^2-9)^{n-1}.\ 2x$

$=(x^2-9)^{n-1}[2(x^2-9)(x+1)+2nx(x^2+2x+3)]$

$=(x+3)^{n-1}(x-3)^{n-1}.q(x)$

Derivative must change sign at x = 3

n – 1 = odd

n = even

n = 6

17. $\vec{u}\times\vec{v}=\vec{w}\quad \vec{u}\perp\vec{v}\quad \vec{u}\perp\vec{w}$

$\vec{v}\times\vec{w}=\vec{u}\qquad \vec{v}\perp\vec{w}$

$$\begin{vmatrix}-t & 1 & 1\\ 1 & -t & 1\\ 1 & 1 & -t\end{vmatrix}=0$$

$\vec{w}=\hat{i}+\hat{j}-2\hat{k}$

$\vec{u}=\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$

$\vec{u}.\vec{w}=0=\alpha+\beta-2\gamma$ t = -1 or 2

$(\vec{v}.\vec{w})\times\vec{v}=\vec{u}\times\vec{v}$ t = -1

$\vec{w}|\vec{v}|^2-0=\vec{w}$ $\alpha+\beta+\gamma=0$

$|\vec{v}|=1(P)$ t = 2

$\alpha=\beta=\gamma$

Also $\vec{u}\times\vec{v}=\vec{w}$

$|\vec{u}||\vec{v}|=|\vec{w}|$

$\Rightarrow|\vec{u}|=|\vec{w}|=\sqrt{6}$

Case – 1

t = 2

$\alpha=\beta=\gamma$

$\alpha^2+\beta^2+\gamma^2=6$

$\alpha=\sqrt{2},-\sqrt{2}$

(S) t + 3 = 5

Case – 2

t = -1

$\alpha+\beta+\gamma=0$

$\alpha+\beta-2\gamma=0$

$\gamma=0,\alpha=-\beta$

(Q) $\alpha=\sqrt{3},\gamma^2=0,\beta=-\sqrt{3}$

(R) $\alpha=\sqrt{3}(\gamma+\beta)^2=\beta^2=3$

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PHYSICS

18. In case of interference,

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

(a) for I to be maximum and minimum cos ϕ is 1 and -1 respectively , i. e.,

$$I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$$
$$= (\sqrt{I_1} + \sqrt{I_2})^2$$

and

$$I_{max} = I_1 + I_2 - 2\sqrt{I_1 I_2}$$
$$= \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$$

According to given problem,

$$\frac{I_{max}}{I_{min}} = \frac{\left(\sqrt{I_1} + \sqrt{I_2}\right)^2}{\left(\sqrt{I_1} - \sqrt{I_2}\right)^2} = \frac{9}{1},$$

i.e., $\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_2} - \sqrt{I_1}} = \frac{3}{1}$

By componendo and dividend,

i.e., $\frac{\sqrt{I_1}}{\sqrt{I_2}} = \frac{3+1}{3-1}$

$$\frac{I_1}{I_2} = \frac{4}{1} = 4$$

(b) Now as for a wave $I \propto A^2$

$$\frac{I_1}{I_2} = \left[\frac{A_1}{A_2}\right]^2, \left[\frac{A_1}{A_2}\right]^2 = 4$$

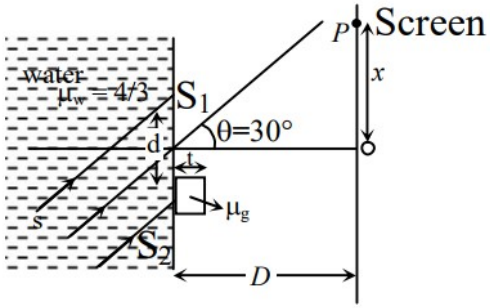
i.e., $\frac{A_1}{A_2} = 2$

19. If central maxima is formed at point P on the screen, then

Path difference = $S_2P + (\mu_g - 1)t - S_1P - \mu_w d\sin 30^o = 0$

$$\Rightarrow S_2P - S_1P = \frac{xd}{D} = \mu_w d\sin 30^o - (\mu_g - 1)\times t$$

$$\Rightarrow x = -1.66 \text{ cm}$$



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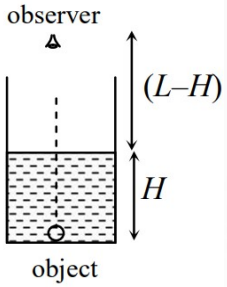
20. $X_{app} = L - H + \dfrac{H}{\mu}$

$\dfrac{dX_{app}}{dt} = \dfrac{dH}{dt}\left(\dfrac{1-\mu}{\mu}\right)$

$\pi r^2 H = V$

$\therefore \dfrac{dH}{dt} = -\dfrac{2H}{r}\dfrac{dr}{dt} = \dfrac{2KH}{r}$

$\therefore \dfrac{dX_{app}}{dt} = \dfrac{2KV}{\pi r^3}\left(\dfrac{(1-\mu)}{\mu}\right)$



21. $a = v\dfrac{dv}{dx} = \dfrac{25}{(x+2)^3}, \dfrac{v^2}{2} = 25\times\left[-\dfrac{1}{2(x+2)^2}\right]_o^x, v^2 = 25\left[\dfrac{1}{4} - \dfrac{1}{(x+2)^2}\right]$

$v = \sqrt{25\left[\dfrac{1}{4} - \dfrac{1}{(x+2)^2}\right]}, v_{\max} = \dfrac{5}{2} = 2.5m/s(at\ x = \infty)$

22.

23.

24.

25.

$1\ell = 10^3(cc)^3$

$1\ell = 10^{-3}\ m^3$

Rate of water flowing $= \dfrac{1000}{7}\ell/\min$

$= \dfrac{1}{7}m^3/\min\ \{\rho = 1000\ kg/m^3\}$

Rate of water flow $= \dfrac{1000}{7}kg/\min$

Power of source $= P \times 10^6$ watt/sec

Let for a 60 sec power supply

Energy of source $= P \times 106 \times 60$

using energy conservation

$P \times 106 \times 60$

$= \dfrac{1000}{7} \times 1 \times 4200\,(60) + \dfrac{1000}{7} \times 1 \times 2268 \times 10^3$

$P \times 60 = \dfrac{42 \times 6}{7} + \dfrac{2268}{7} = 36 + 324$

$P \times 60 = 360$

$P = 6$

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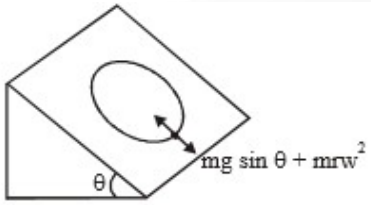
26.

$$P = 700\times10^{3}\times1.6\times10^{-19}\times\frac{dn}{dt} = 10\times10^{-3}$$

$$\frac{dN}{dt} = \frac{10^{-1}}{10^{-14}}\times\frac{1}{7\times16} = \frac{10^{12}}{11.2} = \lambda N_0$$

$$\lambda = \frac{\ln 2}{14\times86400} \Rightarrow N_0 = \frac{14\times86400\times10^{12}}{11.2\ln 2} = 154\times10^{15}$$

27. $f \geq mg\sin\theta + mrw^2$

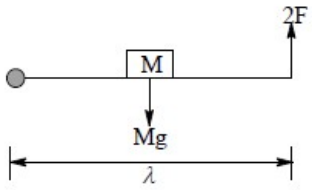


$$\mu mg\cos\theta \geq mg\sin\theta + mrw^2$$

OR $\mu \geq \tan\theta + \frac{w^2 r}{g\cos\theta}$

28. Consider the motions of the bodies both in the original frame of reference and in another frame that moves with velocity V_0 relative to the first one. Apply the Relative motion formulae. If Ann were sitting at the centre of the disc, she would see the whole world around her rotating with the same angular speed ω, but in the opposite direction. That means she would observe Bob standing 12 m away from the centre of the disc, but moving with a speed of $1/6\times12 = 2ms^{-1}$ in a direction perpendicular to the line joining him to the centre of the disc. Although Ann is not sitting at the centre of the disc, but at its edge, the same conclusion applies.

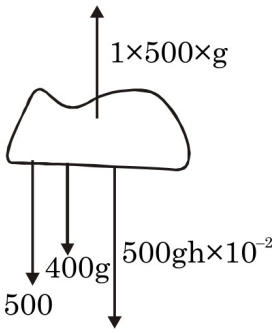
29.



For equilibrium $\left|Mg\frac{\lambda}{2}\right| = |2F|1F = \frac{mg}{4} = \frac{100}{4} = 25N$

30.

31.

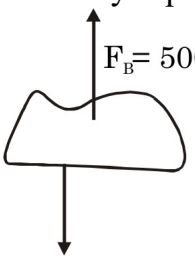


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$500g = 500 + 500gh \times 10^{-2} + 4000$

$500g = 50h \Rightarrow h = 10m$

Force by liquid on upper object



$500gh \times 10^{-2} = 500$ N

$5000 - 500 = 4500N$

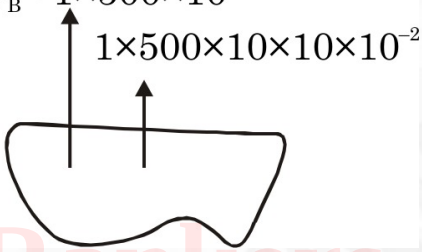
Force by liquid on lower object

$500gh \times 10^{-2} = 500\ N$

$5000 - 50 = 4500\ N$

Force by liquid on lower object

$F_B = 1 \times 500 \times 10$

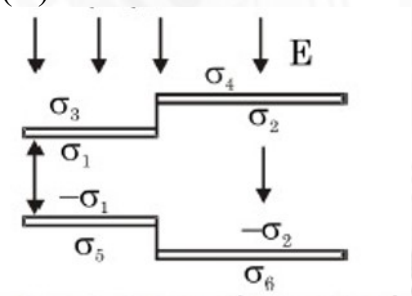


$5000 + 500 = 5500$

Total force by liquid on both object

$5500 + 4500 = 10^4\ N$

32. (A)



$\sigma_1 A_1 + \sigma_2 A_2 = -(\sigma_3 A_1 + \sigma_4 A_2)$

$\sigma_1 A_1 + \sigma_2 A_2 = (\sigma_5 A_1 + \sigma_6 A_2)$

$2(\sigma_1 A_1 + \sigma_2 A_2) = (\sigma_5 - \sigma_3)A_1 + (\sigma_6 - \sigma_4)A_2$

$\frac{\sigma_1}{\epsilon_0} = E_1, \frac{\sigma_2}{\epsilon_0} = E_2, \frac{(\sigma_5 - \sigma_3)}{2\epsilon_0} = E, \frac{(\sigma_6 - \sigma_4)}{2\epsilon_0} = E$

$2A_1E_1\epsilon_0 + 2A_2E_2\epsilon_0 = 2\epsilon_0 EA_1 + 2\epsilon_0 EA_2$

$A_1E_1 + A_2E_2 = E(A_1 + A_2)$ ..... (1)

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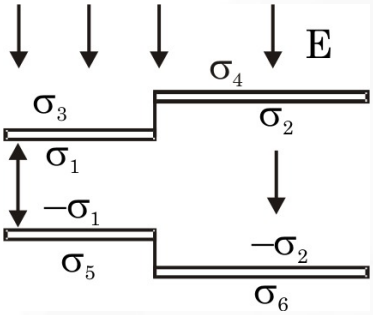
$E_1 d_1 = E_2 d_2$ ..... (2)

$$E_1 = Ed_2 \frac{(A_1 + A_2)}{A_1 d_2 + A_2 d_1} = \frac{100 \times 2(1+2)}{(2+2\times 1)} = \frac{200 \times 3}{4} = 150$$

$$E_2 = Ed_1 \frac{(A_1 + A_2)}{(A_1 d_2 + A_2 d_1)} = 75$$

$E_1 + E_2 = 225$

$E_1 - E_2 = 75$



33. $R_1 = 2\rho \cdot \frac{2\ell}{\pi r^2} = 4\rho \cdot \frac{\ell}{A}$

$$R_2 = \rho \cdot \frac{2\ell}{\pi(3r)^2} = \frac{2}{9}\rho \cdot \frac{\ell}{A}$$

$$R_3 = 3\rho \cdot \frac{3\ell}{\pi(2r)^2} = \frac{9}{4}\rho \cdot \frac{\ell}{A}$$

$$\frac{P_2P_3}{P_1^2} = \frac{R_2R_3}{(R_1)^3} = \frac{\frac{2}{9}\times\frac{9}{4}}{16} = \frac{1}{32}$$

$E = \rho J$

$J = nev_d = \frac{I}{A}$

$E = \frac{\rho \cdot I}{A}$

$E_1 = \frac{\rho \cdot I}{\pi r^2} = \frac{2\rho I}{A}$

$E_2 = \frac{\rho . I}{\pi(3r)^2} = \frac{\rho I}{9A}$

$E_3 = \frac{3\rho . I}{\pi(2r)^2} = \frac{3\rho I}{4A}$

$$\frac{E_1E_2}{E_3^2} = \frac{2\times\frac{1}{9}}{\frac{9}{16}} = \frac{32}{81}$$

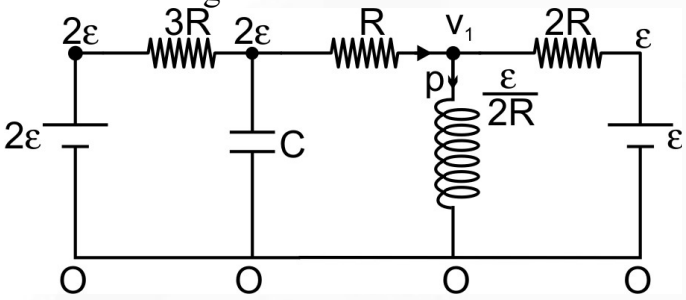
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Similarly value of

$$\frac{\upsilon d_2 \upsilon d_3}{\upsilon d_1^2} = \frac{1}{4}$$

$$\frac{V_2 \cdot V_3}{V_1^2} = \frac{1}{32}$$

34. Just after S_3 and S_4 are closed current in inductor is unchanged and charge on capacitor is also unchanged



KCL at junction P

$$\frac{v_1 - 2\varepsilon}{R} + \frac{v_1 - 2\varepsilon}{2R} + \frac{\varepsilon}{2R} = 0$$

$$v_1 = \frac{4\varepsilon}{3}$$

So, current in resistance R is

$$I_1 = \frac{2\varepsilon - \frac{4\varepsilon}{3}}{R} = \frac{2\varepsilon}{3R} = 2.5A$$

Similarly, $I_2 = \frac{2\varepsilon}{3R} = -2.5A$

Current through capacitor = 2.5 A

So, u $= \frac{Q^2}{2C}$ $\frac{dU}{dt} = \frac{1}{2C} \cdot 2Q \cdot \frac{dQ}{dt}$ $= \frac{1}{2C} \cdot 2 \cdot Cv \cdot \frac{dQ}{dt}$

$$= v \cdot \frac{dQ}{dt} = 2\varepsilon(2.5) = 20 \times 2.5 = 50$$

As $\frac{dQ}{dt}$ is –ve So, $\frac{du}{dt}$ is also negative

In inductor, $L\frac{di}{dt} = \frac{4\varepsilon}{3}; \frac{di}{dt} = \frac{4\varepsilon}{3}$

B = $\mu_0 ni$ $\frac{dB}{dt} = \mu_0 n \frac{4\varepsilon}{3L} L = \mu_0 n^2 \pi r^2 \ell$

$$\frac{dB}{dt} = \mu_0 n \frac{4\varepsilon}{3.\mu_0.n^2 \pi r^2 \ell} \quad \frac{dB}{dt} = \frac{dB}{dt} = \frac{4}{3} \times \frac{10}{16} \times 3 = 2.5$$

As, $\frac{di}{dt}$ is positive, So, $\frac{dB}{dt}$ is also positive.

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CHEMISTRY

35. (B)

$$Z = \frac{PV}{RT}$$

$$P = \frac{RT}{V-b} - \frac{a}{V^2} \qquad Z = \frac{PV}{RT} = \frac{V}{V-b} - \frac{a}{VRT}$$

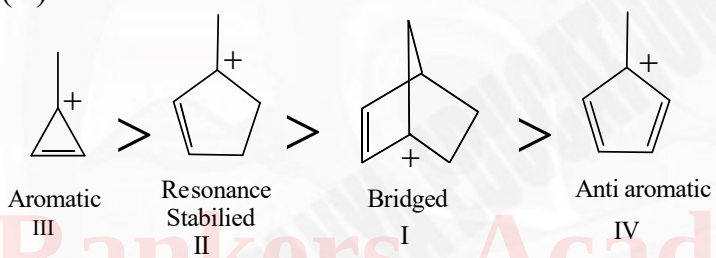
$$T_b = \frac{a}{Rb} \qquad Z = \frac{V}{V-b} - \frac{a}{VR\frac{a}{Rb}}$$

$$Z = \frac{V}{V-b} - \frac{b}{V} = 1 + \frac{b^2}{V(V-b)} \qquad Z = \left(1 - \frac{b}{v}\right)^{-1} - \frac{b}{V}$$

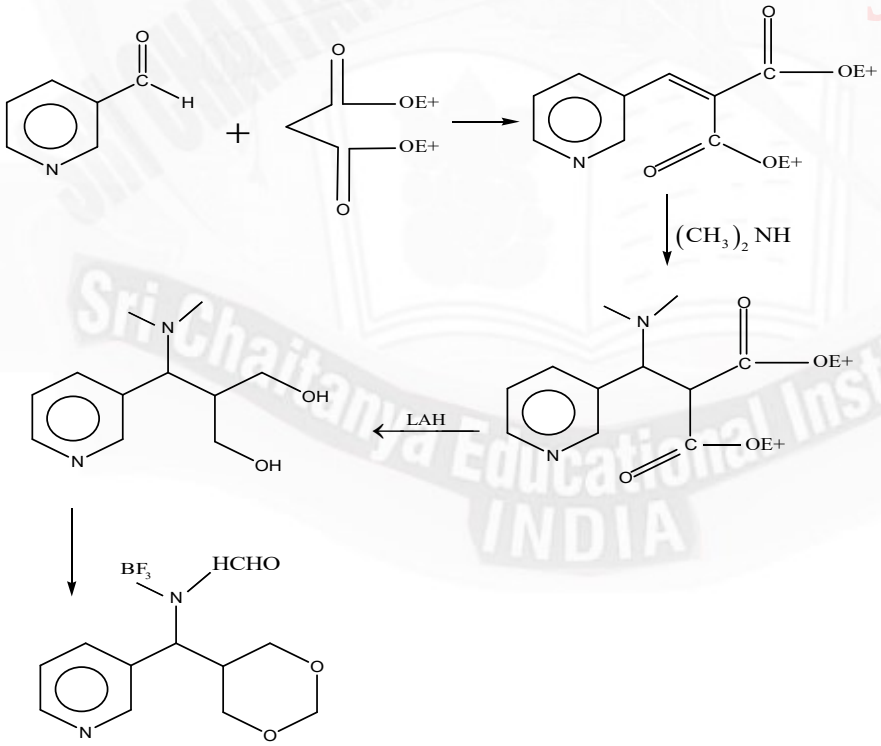
$$Z = 1 + \frac{b}{v} + \frac{b^2}{v^2} + \frac{b^3}{v^3} - - - - - - - - \frac{b}{v}$$

$$Z = 1 + \frac{b^2}{v^2} + \frac{b^3}{v^3} - - - - - - -$$

36. (C)



37.



38. Ag^+ displaced to interstitial sites $= N \times \frac{2}{100} = 0.02N$

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Ag^+ missing due to schottky defect

$= N \times \frac{0.5}{100} = 0.005N$

Br^- missing due to schottky defect

$= N \times \frac{0.5}{100} = 0.005N$

No ,of Ag^+ at lattice sites

$= N - 0.02N - 0.005N = 0.975N$

Total number of Br^- = N-0.005N = 0.995N

$\frac{\rho}{\rho_0} = \frac{mass\ of\ defective\ crystal}{mass\ of\ perfect\ crystal}$

$= \frac{0.995N}{N} = 0.995$

Total number of tetrahedral voids = 2N

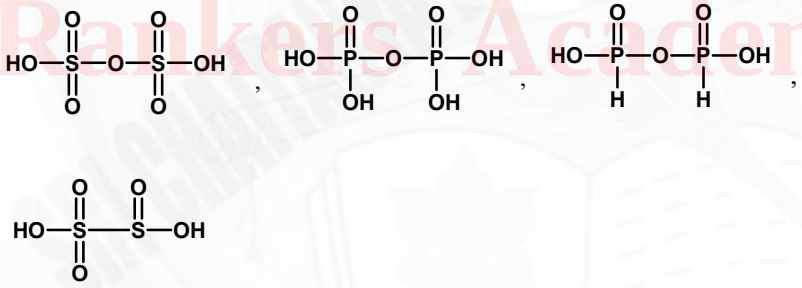
Fraction of tetrahedral voids occupied

$= \frac{0.02N}{2N} = 0.01$

Coordination number of tetrahedral void = 4

39. ΔS_{total} is positive for micelle formation due to increase in entropy of solvent.

40.



41. $2PbO.PbO_2 + HNO_3 \rightarrow Pb(NO_3)_2 + PbO_2 + H_2O$

$PbO + HNO_3 \rightarrow Pb(NO_3)_2 + H_2O$

PbO_2 is filtered out as such

so, Here HNO_3 is behaving as acid only and not as oxidising agent

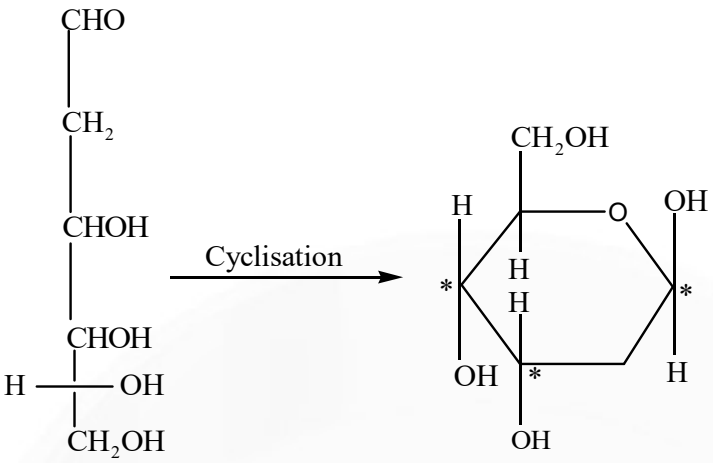
(B) $POCl_3 + SOCl_2$

(C) $C + H_2O$

(D) $XeO_3 + HF$

42. The D-form of given sugar is

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D-Sugar D-Pyranose

Configuration at the chiral carbons(starred) can be changed maintaining D- configuration.
Hence, the total number of stereoisomers of D-pyranose = $2^3 = 8$
Thus, the correct integer is 8.

43. Graphite, $(BN)_x, B_3N_3H_6, C_6H_6$

44. $$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.6\times10^{-34}}{\sqrt{2\times9\times10^{-31}\times50\times10^{3}}} = \frac{6.6\times10^{-34}}{\sqrt{900\times10^{-28}}}$$

$$= \frac{6.6\times10^{-34}}{30\times10^{-14}} = 0.22\times10^{-20} = 0.22\times10^{-z} \quad \therefore z = 20$$

45. $t = 0 \quad \frac{P_0 - P_s}{P_0} = \frac{2}{2+12} = \frac{1}{7};\ P_s = \frac{6P_0}{7}$

$$t = 10 \quad \frac{P^0 - \frac{4P^0}{5}}{P^0} = \frac{2+2x}{2+2x+12}$$

$x = 0.5$

$A \quad \rightarrow 2B + C$

$2 - 0.5 \quad 1 \quad 0.5$

$$t = 20 \quad \frac{P^0 - \frac{3P^0}{4}}{P^0} = \frac{2+2x}{2+2x+12}$$

x = 1

$A \quad \rightarrow 2B + C$

$2 - 1 \quad 2 \quad 1$

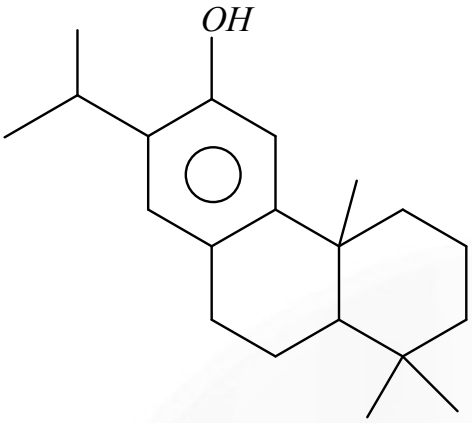
So reaction follows zero order kinetics.

x = kt

$0.5 = k \times 10$

k = $0.05\, mol\, \min^{-1}$

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46. $C_{20}H_{30}O$

47. $X \Rightarrow (NH_4)_3PO_4 \cdot 12MoO_3$

15th group and sp^3 hybridized atoms in X = 4, (3, N-atoms + 1-atom)

48. **(I)** $CoCl_2 + KNO_2 + CH_3COOH \rightarrow$

$K_3[Co(NO_2)_6]\downarrow (yellow) + NO + KCl + CH_3COOK + H_2O$

(II) $Zn + 4HNO_3 \rightarrow Zn(NO_3)_2 + 2NO_2 + 2H_2O$

(III) $PbS + O_3 \rightarrow PbSO_4 + 4O_2$

(IV) $NaBH_4 + I_2 \rightarrow B_2H_6 + NaI + H_2$

49. (A) $H_2 + 2Ag^+ \rightarrow 2H^+ + 2Ag$

$E^0_{cell} = 0.80V$

$$E_{cell} = E^0_{cell} - \frac{0.06}{2}\log\frac{[H^+]^2}{[Ag^+]^2}$$

$P^H = 5$

$[H^+] = 10^{-5} M$

$[K_{SP}] = S^2 = 10^{-10} M$

$[Ag^+] = S = 10^{-5} M$

$$E_{cell} = E^o_{cell} - \frac{0.06}{2}\log\left(\frac{10^{-5}}{10^{-5}}\right)^2$$

$E_{cell} = 0.80V$

(B) $$E = 0 - \frac{0.06}{2}\log\frac{[H^+]}{P_{H_2}} = -0.06\log[H^+]$$

$= 0.06 \times 5 = 0.30V$

(C) $HA + NaOH \rightarrow NaA + H_2O$

0.1 0.05 0.05 0 0.15

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$$K_a \frac{[H^+][A^-]}{[HA]}$$

$$[H^+] = \frac{10^{-5} \times 0.05}{0.15} = \frac{1}{3} \times 10^{-5} M$$

$$E_{cell} = 0.80 - 0.06 \log \frac{10^{-5}}{3 \times 10^{-5}}$$

$= 0.80 + 0.06 \log 3 = 0.83\text{V}$

(D) $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$

$Ag^+ + 2NH_3 \rightleftharpoons [Ag(NH_3)_2]^+ \; K_f = 10^8$

S 2

x 2-2s S

$$K_{SP} \times K_f = x \times S \times \frac{S}{(2-2S)^2 \times x} \qquad = \left(\frac{S}{2-2S}\right)^2 = 10^{-2}$$

$$\frac{S}{2-2S} = 0.1 \Rightarrow S = 0.2 - 0.2S$$

$1.2S = 0.2 \qquad S = \frac{0.2}{1.2} = \frac{1}{6} M$

$$K_{sp} = x \times \frac{1}{6} = 10^{-10} \qquad x = 6 \times 10^{-10}$$

$$E_{cell} = 0.80 - 0.06 \log \frac{10^{-5}}{6 \times 10^{-10}}$$

$$= 0.80 - 0.06 \log \frac{10^{-5}}{6} \qquad = 0.80 - 0.06 \times (5 - \log 6)$$

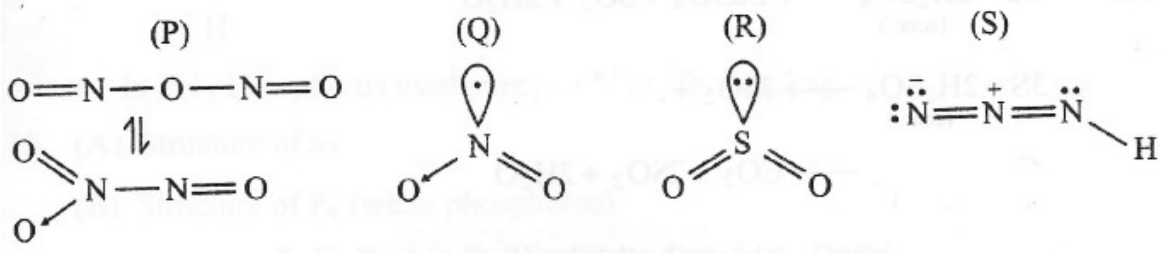
$= 0.80 - 0.06 \times 4.2 = 0.548$ V

50. $\Delta H_f^0(CO_2)$: Q $C(\text{grap}) + O_2(g) \rightarrow CO_2(g)$

$\Delta H_f^0(H_2O(g))$: S $H_2(g) + \frac{1}{2} O_2(g) \rightarrow H_2O(g)$

$\Delta H_{atom}^0(H_2O, \ell)$: R $H_2O(d) \rightarrow 2H(g) + O(g)$

51.



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